



△ ANCIENT PLATEAU

This plateau in Venezuela is one of hundreds of dramatic tabletop mountains, or tepuis, formed from the remains of a sandstone plateau that built up on top of the ancient Guiana Shield.

THE SHAPING OF CENTRAL AND SOUTH AMERICA

Attached to Central America, South America is a huge continent—with high plateaus of old rocks in the east and the towering Andes to the west. It became an independent continent about 90 million years ago, when it split away from Africa.

Shields and tablelands

To the east, a large portion of South America's land is dominated by three giant slabs of ancient rock, typically made of gneiss and granite, forming the Patagonian Shield, the Brazilian Shield, and the Guiana Shield. These slabs survived and moved together as a group for billions of years, even when South America was part of the southern supercontinent Gondwana. South America, along with Africa, split from Gondwana about 180–170 million years ago. Then, 140–90 million years ago, South America slowly split from Africa and the shields eventually became the bedrock around which the rest of South America has been gradually built.

Today, the shields form a base under some of the world's largest high plateaus. In places, these old plateaus, or tablelands, end so abruptly in high cliffs that people once believed they were lost worlds where ancient creatures might survive.

The world's highest waterfall, Angel Falls, plunges over the edge of the Guiana Shield, dropping 3,212 ft (979 m). The slabs are by no means intact. In places, they have cracked and slipped to open up huge valleys—for example, the Parana, Sergipe, and São Francisco rivers now flow through such valleys in the Brazilian Shield.

The rise of the Andes

After it separated from Africa and became a separate continent, South America drifted slowly west, allowing the South Atlantic to expand. Its relentless movement brought it into direct collision with the Nazca Plate under the Pacific Ocean. The collision forced the eastward-moving, smaller but heavier Nazca Plate to subduct under the oncoming, larger but lighter South American Plate. As the

Central and South America have only been linked by land for 3 million years

KEY EVENTS

80–70 million years ago Complex tectonic activity in a shallow ocean north of South America causes uplifting magma to cool in an arc of new islands—forming the Caribbean.

50 million years ago The Nazca Plate under the Pacific Ocean subducts beneath the South America Plate to the west of the continent, and this causes the Andes to start rising. Most of the mountains have grown over the last 10 million years.

15–10 million years ago The Andes rise even higher and block the Amazon River from flowing into the Pacific. It flows eastward and pools into a lake. About 10 million years ago, the Amazon starts to flow into the Atlantic, lowering the lake's water level, and the lush Amazon Rain Forest grows in its place (see opposite page).

180 million years ago The supercontinent Gondwana comprises South America, Africa, India, Australia, and Antarctica. South America and Africa separate from the rest about 180–170 million years ago, and South America starts its own journey when it splits from Africa 140–90 million years ago.

16–13 million years ago The area that will become the Andean Altiplano undergoes its first bout of elevation, with two more instances occurring 13–9 and 10–6 million years ago, which bring the area closer to its present-day height.

14 million years ago Desertification starts to occur in the area that will form the Atacama Desert.

3 million years ago A land bridge called the Isthmus of Panama forms, connecting South America with North America.